



VAIBHAV SINGH

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B.Tech, Computer Science and Engineering

Address : Boys Hostel 2, SRM Ramapuram, Ramapuram, Bharathi Salai, Chennai, Tamil Nadu



Summary

Computer Science Engineering student focused on hands-on systems programming projects, low-level emulators, backendservices, and full-stack applications using Java, Spring Boot, and React. Strong interest in runtime environments,performance optimization, and scalable RESTful APIs.

Education

SRM IST-Ramapuram

B.Tech · Computer Science and Engineering
2023 - 2027

2027

CGPA - 9.24/10

St. Paul's College

Class XII - ICSE · Mpc · Lucknow, Uttar Pradesh

2022

Percentage - 76.4%

St. Paul's College

Class X - ICSE · Lucknow, Uttar Pradesh

2020

Percentage - 91.8%

Experience

INTELLECTS SRMIST Ramapuram

· Member

Aug 2024 - Present

Chennai, Tamil Nadu

- Collaborated in a cross-functional team to architect 3 modular backend microservices using Java and Spring Boot, optimizing database indexing to reduce API response overhead by 25%.
- Spearheaded a campus competitive programming initiative, growing an active community to 100+ students.
- Conducted technical software development seminars for online campus events reaching 65+ attendees.

Projects

NES Emulator | [mrborn2suffer/NES-or-Mess](#)

Dec 2025 - Jun 2026

· Java, Assembly, MOS 6502 Architecture

Systems Programming / Emulation

· Engineered a cycle-accurate MOS 6502 CPU emulator handling 56 base instructions over a 256-opcode matrix at 1.79 MHz.

· Developed a Picture Processing Unit (PPU) executing stable 60 FPS scanline rendering with sub-pixel scrolling.

· Integrated memory bank switching (NROM, MMC1, MMC3, AxCROM) ensuring compatibility with 500+ commercial ROMs.

Aurora | [mrborn2suffer/Aurora](#)

Jan 2026 - Jun 2026

· Spring Boot, React, JWT, Relational DB

Full-Stack Web Development

· Created a free digital reading application using Spring Boot and React featuring secure token-based JWT authentication.

· Designed a responsive layout optimized with content carousels and multi-criteria dynamic sorting features.

· Implemented 15+ RESTful API endpoints coupled with a relational database layer to handle user profiling and book discovery loops.

3D Raycasting Engine | [mrborn2suffer/Wolfstein](#)

Feb 2026 - Present

· Java, DDA Algorithm, Vector Math

Graphics & Game Development

· Utilizing the Digital Differential Analyzer (DDA) algorithm, coded a 3D graphics rendering engine from scratch for real-time grid ray projections.

· Applied linear algebra and vector mathematics to compute ray-wall intersections, resolving 100% of camera lens distortion across a 60-degree Field of View.

· Optimized the core execution loop, maintaining 60+ FPS using custom delta-time driven collision logic.

Certifications

Oracle Certification Oracle Oracle · 102924502OCI25DOPOCP Oracle Certification for DevOps	Jun 2025
NPTEL Certification NPTEL NPTEL · NPTEL24CS96S156301331 NPTEL Certification for Data Structure and Algorithms in Java	Oct 2024
SIH 2025 SIH2025 SRMIST RAMAPURAM Smart India Hackathon (SIH 2025) Internal Nominee	Sep 2025
AWS Training & Certification AWS Certificate AWS Foundation AWS Training Certification for Machine Learning Basics	Aug 2025

Volunteering

SGPGIMS, Lucknow Volunteer · Community Connect Successfully contributed 5 weeks of service to the Department of Immunology, SGPGIMS, Lucknow	May 2025 - Jun 2025
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Competitions

SIH 2025 Hackathon Link Top 50 · Hardware and Software hackathon
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Skills

Programming Languages: Java , Python
Languages and Frameworks: Spring Boot, Java Web Toolkit , LibGDX
Tools and Technologies: Git , Firebase, MongoDB, Maven , Docker
Areas of Interest: Object-oriented Programming , Problem Solving, Backend Development , REST APIs

Languages

English [Professional Working Proficiency], Hindi [Professional Working Proficiency]
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Links

LinkedIn , GitHub , CodeChef , LeetCode
